

# ACC Two-Channel Torque Assembly

## Physical Conditions — Smoothstep Gate Inputs

Proximity  $|\Delta x|$  · Quietness  $\text{EMA}(|v|)$  · Stability  $\theta, \dot{\theta}$  · Contact  $F_z^L, F_z^R$  ·  
Terrain  $h^L_{\text{gnd}}, h^R_{\text{gnd}}$

### Wheel Torque $\tau_w$

$\tau_{\text{balance}}$   
Sagittal Balance Core  
(pitch PD + vel. damp + pos-P  
+ lateral + yaw)  
always active

+

$g_{\text{anchor}} \cdot \tau_{\text{anchor}}$   
Proximity-Gated Anchor  
(integral + damping boost)  
gated by  $g_{\text{prox}}, g_{\text{env}}$

+

$g_{\text{flight}} \cdot \tau_{\text{flight}}$   
Contact-Loss Recovery  
(reaction-wheel attitude PD)  
gated by  $g_{\text{flight}}$



$$\tau_w = \tau_{\text{bal}} + g_{\text{anc}} \cdot \tau_{\text{anc}} + g_{\text{flt}} \cdot \tau_{\text{flt}}$$

### Leg-Joint Torque $\tau_q$

$\tau_{\text{posture}}(h_{\text{cmd}})$   
Posture & Yaw Stability  
(Jacobian PD + hip-yaw  
divergence + homing)  
always active, height-scheduled

+

$g_{\text{terrain}} \cdot \Delta \tau_{\text{posture}}$   
Per-Leg Ground Adaptation  
(split height commands  
+ leveling integrator)  
gated by  $g_{\text{terrain}}$



$$\tau_q = \tau_{\text{post}} + g_{\text{terr}} \cdot \Delta \tau_{\text{post}}$$

### Smoothstep Gate Definitions

$g_{\text{prox}} = 1 - \text{ss}(|\Delta x|; 0.05 \text{ m}, 0.15 \text{ m})$  |  $g_{\text{env}} = 1 - \text{ss}(\text{EMA}(|v_{\text{sag}}|); 0.25 \text{ m/s}, 0.50 \text{ m/s})$

Asymmetric EMA on  $|v_{\text{sag}}|$ :  $\alpha_{\text{attack}} = 0.35$  ( $\tau \approx 23 \text{ ms}$ ) |  $\alpha_{\text{release}} = 0.0067$   
( $\tau \approx 1.5 \text{ s}$ ) | pitch stiffness fixed at  $k_p = 50$

$g_{\text{flight}}$ :  $F_z < 0.5 \text{ mg}$  for  $\geq 2$  steps, release  $\geq 5$  steps + 150 ms ramp |  
 $g_{\text{terrain}}$ : per-leg ground estimate, frozen while unloaded (seek on 3 mm drop-below); leveling trim on  $|\Delta h| \geq 2 \text{ cm}$

Both channels assembled independently  $\rightarrow$  final 10-DoF torque command

